

Quantum Optics — Lecture 2

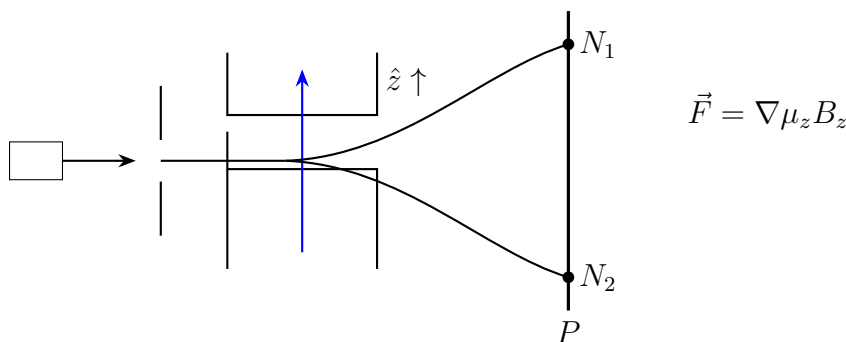
Two-level atomic systems

Notes of 15 September 2026

1 Two level atomic systems

1.1 Stern–Gerlach and spin $\frac{1}{2}$

$$\langle \vec{\mu} \rangle = -g \mu_B \left\langle \frac{\vec{J}}{\hbar} \right\rangle$$



Interaction energy of the magnetic moment with the magnetic field is just $-\vec{\mu} \cdot \vec{B}$.

$$F_z = \frac{\partial}{\partial z} (\vec{\mu} \cdot \vec{B}) \simeq \mu_z \frac{\partial B_z}{\partial z}$$

- If $\mu_z > 0$ ($S_z < 0$) the atom experiences an upward force.
- If $\mu_z < 0$ ($S_z > 0$) the atom experiences a downward force.
- The atoms in the furnace are randomly oriented. So classically we would expect values between $|\mu|$ and $-|\mu|$; this would lead to a spread in the values, unlike the only two spots being observed.

$\Rightarrow$ Later this was realized to be spin.

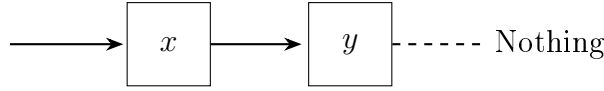
$\Rightarrow$ We also see the weird measurement effects with multiple Stern–Gerlach experiments.

$$\vec{E} = E_0 \hat{x} \cos(kz - \omega t)$$

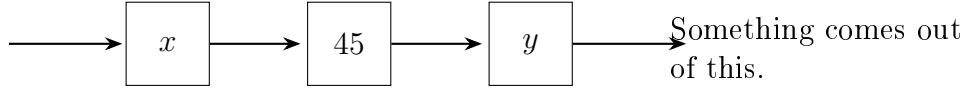
Likewise we may consider y -polarized light, also propagating in the z -direction,

$$\vec{E} = E_0 \hat{y} \cos(kz - \omega t).$$

Then we have that



but



1.2 Constructing spin operators from sequential SG experiments

We know we can write

$$S_z = \frac{\hbar}{2} \left[|+\rangle \langle +| - |- \rangle \langle -| \right].$$

Classically a current loop has the magnetic moment

$$\begin{aligned} \mu &= \frac{IA}{c} = \frac{qA}{Tc} = \frac{qvA}{2\pi r c} = \frac{qv \pi r^2}{2\pi r c} \\ &= \frac{qv}{2c} r = \frac{q}{2mc} mvr = \frac{q}{2mc} L, \end{aligned}$$

so

$$\mu = \frac{q}{2mc} L.$$

For spin $\mu = g \frac{q}{2mc} S$; $g = 1 \Rightarrow$ classical orbital value. Dirac says $g = 2$:

$$\mu = \frac{q}{mc} S, \quad |S_z| = \frac{m_e c}{e} |\mu_z|, \quad \mu_z = \frac{e\hbar}{2m_e c} (\text{exp.}) \quad \Rightarrow \quad S_z = \frac{\hbar}{2}.$$

We have that

$$S_z = \frac{\hbar}{2} \left[|+\rangle \langle +| - |- \rangle \langle -| \right]$$

Observation. The S_x+ beam splits 50/50 in $SG\hat{z}$, so we require that

$$|\langle +|S_x; + \rangle|^2 = |\langle -|S_x; + \rangle|^2 = \frac{1}{2},$$

but it also must be the case that

$$|S_x; + \rangle = c_1 |+\rangle + c_2 |- \rangle.$$

Thus combining the two and taking into account an unphysical phase we have that

$$|S_x; + \rangle = \frac{1}{\sqrt{2}} |+\rangle + \frac{e^{i\delta_1}}{\sqrt{2}} |- \rangle.$$

Then since $|S_x; - \rangle$ is necessarily orthogonal to $|S_x; + \rangle$ we have that

$$|S_x; - \rangle = \frac{1}{\sqrt{2}} |+\rangle - \frac{e^{i\delta_1}}{\sqrt{2}} |- \rangle.$$

Observation. S_y also splits in SGx , so

$$\begin{aligned}\langle S_y; + | S_x; + \rangle &= \frac{1}{2}(1 + e^{i(\delta_1 - \delta_2)}), \\ |\langle S_y; + | S_x; + \rangle|^2 &= \frac{1}{4}|1 + e^{i(\delta_1 - \delta_2)}|^2 = \frac{1}{2}(1 + \cos(\delta_1 - \delta_2)) = \frac{1}{2} \\ &\iff \cos(\delta_1 - \delta_2) = 0 \iff \delta_2 - \delta_1 = \pm \frac{\pi}{2}.\end{aligned}$$

Let $\delta = \frac{\pi}{2}$, so

$$|S_x; \pm\rangle = \frac{1}{\sqrt{2}}(|+\rangle \pm |-\rangle) \quad \text{and} \quad |S_y; \pm\rangle = \frac{1}{\sqrt{2}}(|+\rangle \pm i|-\rangle).$$

So

$$\begin{aligned}S_x &= \frac{\hbar}{2}(|S_x; +\rangle \langle S_x; +| - |S_x; -\rangle \langle S_x; -|) = \frac{\hbar}{2}(|+\rangle \langle -| + |-\rangle \langle +|), \\ S_y &= \frac{\hbar}{2}(-i|+\rangle \langle -| + i|-\rangle \langle +|).\end{aligned}$$

2 General two level systems

A quantum system whose Hilbert space is exactly $\mathbb{C}^2$. They have two basis states denoted by

$$|\uparrow\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad |\downarrow\rangle = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The general state is

$$|\psi\rangle = c_0 |0\rangle + c_1 |1\rangle, \quad |c_0|^2 + |c_1|^2 = 1, \quad c_0, c_1 \in \mathbb{C}$$

$\Rightarrow$ 4 real numbers, -1 for normalization, -1 for global phase $\Rightarrow$ 2 independent real parameters.

2.1 Pauli matrices

$$\begin{aligned}\sigma_x &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ \sigma_i^2 &= 1, \quad \sigma_i \sigma_j = \delta_{ij} \mathbb{1} + i\epsilon_{ijk} \sigma_k, \quad [\sigma_i, \sigma_j] = 2i\epsilon_{ijk} \sigma_k, \quad \{\sigma_i, \sigma_j\} = 2\delta_{ij} \mathbb{1}\end{aligned}$$

Observe

$$\begin{aligned}\hat{\sigma} &= |g\rangle \langle e| = \frac{1}{2}(\sigma_x + i\sigma_y) \\ \hat{\sigma}^\dagger &= |e\rangle \langle g| = \frac{1}{2}(\sigma_x - i\sigma_y)\end{aligned}$$

$\Rightarrow \hat{\sigma} + \hat{\sigma}^\dagger = \sigma_x$, and

$$|e\rangle \langle e| = \frac{1}{2}(\mathbb{1} + \sigma_z).$$

Proposition. $\{\mathbb{1}, \sigma_x, \sigma_y, \sigma_z\}$ is a basis for all 2×2 matrices.

Proof. Any Hermitian matrix H can be written as

$$H = \begin{pmatrix} a & c - id \\ c + id & b \end{pmatrix} = \frac{a+b}{2} \mathbb{1} + c \sigma_x + d \sigma_y + \frac{a-b}{2} \sigma_z. \quad \square$$

We can therefore write every 2-level Hamiltonian as

$$\hat{H} = E_0 \mathbb{1} + \frac{\hbar}{2} \vec{\Omega} \cdot \vec{\sigma}$$

for some $E_0 \in \mathbb{R}$.

Note. Suppose $H = a\mathbb{1} + b\sigma_x + c\sigma_y + d\sigma_z$, then

$$H = \begin{bmatrix} H_{00} & H_{01} \\ H_{01}^* & H_{11} \end{bmatrix} \Rightarrow a = \frac{H_{00} + H_{11}}{2}, \quad b = \operatorname{Re} H_{01}, \quad c = -\operatorname{Im} H_{01}, \quad d = \frac{H_{00} - H_{11}}{2},$$

or using $\langle A, B \rangle = \frac{1}{2} \operatorname{Tr}(AB)$ we have

$$a = \frac{1}{2} \operatorname{Tr}(H), \quad b = \frac{1}{2} \operatorname{Tr}(H\sigma_x), \quad \text{etc.}$$

So from this

$$\vec{\Omega} = \frac{2}{\hbar} (b, c, d), \quad E_0 = a,$$

- $\hbar\Omega_z = H_{00} - H_{11}$ (energy splitting),
- $\hbar\Omega_x = 2 \operatorname{Re} H_{01}$,
- $\hbar\Omega_y = -2 \operatorname{Im} H_{01}$ (coupling, magnitude and phase).

2.2 Eigenvalues of a general 2-level Hamiltonian $\hat{H}$

We can make use of the Pauli identities to find the energy levels of any 2-level $\hat{H}$. Start by

$$\vec{\Omega} \cdot \vec{\sigma} = \Omega_x \sigma_x + \Omega_y \sigma_y + \Omega_z \sigma_z = \Omega_i \sigma_i.$$

Then

$$(\vec{\Omega} \cdot \vec{\sigma})^2 = \Omega_i \sigma_i \Omega_j \sigma_j = \Omega_i \Omega_j \sigma_i \sigma_j$$

since Ω_i, Ω_j are scalars. And then using $\sigma_i \sigma_j = \delta_{ij} \mathbb{1} + i\epsilon_{ijk} \sigma_k$ we have

$$\begin{aligned} (\vec{\Omega} \cdot \vec{\sigma})^2 &= \Omega_i \Omega_j \delta_{ij} \mathbb{1} + \underbrace{i \Omega_i \Omega_j \epsilon_{ijk} \sigma_k}_{=0} \\ &= \Omega_i \Omega_i + 0 = |\Omega|^2, \end{aligned}$$

where the second term vanishes because

$$i \Omega_i \Omega_j \epsilon_{ijk} \sigma_k = i \Omega_j \Omega_i \epsilon_{ijk} \sigma_k = i \Omega_i \Omega_j (-\epsilon_{ijk}) \sigma_k = 0.$$

Thus

$$(\vec{\Omega} \cdot \vec{\sigma})^2 = |\Omega|^2 \mathbb{1}$$

Let $M = \vec{\Omega} \cdot \vec{\sigma}$ and if $|v\rangle$ is an eigenvector with eigenvalue λ we have that

$$\begin{aligned} M|v\rangle &= \lambda|v\rangle \Rightarrow M^2|v\rangle = \lambda M|v\rangle = \lambda^2|v\rangle, \\ \text{but } M^2 &= |\Omega|^2 \mathbb{1} \Rightarrow M^2|v\rangle = |\Omega|^2 \mathbb{1}|v\rangle = \lambda^2 \mathbb{1}|v\rangle \\ &\iff |\Omega|^2 = \lambda^2 \iff \boxed{\lambda = \pm|\Omega|}. \end{aligned}$$

Furthermore we have that $\lambda^2 \geq 0$, $\lambda \in \mathbb{R}$. And

$$\text{Tr}(\vec{\Omega} \cdot \vec{\sigma}) = \Omega_i \text{Tr}(\sigma_i) = 0 = \lambda_1 + \lambda_2 \iff \lambda_1 = -\lambda_2,$$

consistent with our λ values.

Then for the two possible energy levels we have

$$\hat{H} |\pm\rangle = \left(E_0 \pm \frac{\hbar}{2} |\vec{\Omega}|\right) |\pm\rangle \iff E_{\pm} = E_0 \pm \frac{\hbar}{2} |\vec{\Omega}|,$$

$$\boxed{E_+ - E_- = \hbar |\vec{\Omega}|}$$

Corollary. For a unit vector $\hat{n}$ we have

$$e^{-i\frac{\theta}{2}\hat{n}\cdot\vec{\sigma}} = \cos\left(\frac{\theta}{2}\right)\mathbb{1} - i\sin\left(\frac{\theta}{2}\right)(\hat{n} \cdot \vec{\sigma}).$$

Proof. We have that

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}, \quad \text{then let} \quad A = -i\frac{\theta}{2} \underbrace{\hat{n} \cdot \vec{\sigma}}_M = -i\frac{\theta}{2} M.$$

Observe $M^0 = 1$, $M^1 = M$, $M^2 = \mathbb{1}$, $M^3 = M^2 M = M$, $M^4 = M^3 M = M^2 = \mathbb{1}$, $M^5 = M^4 M = M$, $\dots$; from this

$$n \text{ odd} \Rightarrow M^n = M, \quad n \text{ even} \Rightarrow M^n = \mathbb{1}.$$

Then

$$\begin{aligned} e^{(-i\frac{\theta}{2})M} &= \sum_{n=0}^{\infty} \left(-\frac{i\theta}{2}\right)^n \frac{1}{n!} M^n \\ &= \sum_{n \text{ even}} \left(-\frac{i\theta}{2}\right)^n \frac{1}{n!} \underbrace{M^n}_{\mathbb{1}} + \sum_{n \text{ odd}} \left(-\frac{i\theta}{2}\right)^n \frac{1}{n!} \underbrace{M^n}_M \\ &= \sum_{k=0}^{\infty} \left(-\frac{i\theta}{2}\right)^{2k} \frac{1}{(2k)!} + \sum_{k=0}^{\infty} \left(-\frac{i\theta}{2}\right)^{2k+1} \frac{M}{(2k+1)!} \\ &= \sum_{k=0}^{\infty} (-1)^k \left(\frac{\theta}{2}\right)^{2k} \frac{1}{(2k)!} - i \sum_{k=0}^{\infty} (-1)^k \left(\frac{\theta}{2}\right)^{2k+1} \frac{1}{(2k+1)!} M \\ &= \cos\left(\frac{\theta}{2}\right)\mathbb{1} - i\sin\left(\frac{\theta}{2}\right)M. \end{aligned} \quad \square$$

Thus we have that

$$e^{-i\frac{\theta}{2}\hat{n}\cdot\vec{\sigma}} = \cos\left(\frac{\theta}{2}\right)\mathbb{1} - i\sin\left(\frac{\theta}{2}\right)\hat{n} \cdot \vec{\sigma}.$$

For

$$\hat{U}(t) = e^{-i\hat{H}t/\hbar} = e^{-iE_0 t/\hbar} e^{-i\frac{|\vec{\Omega}|}{2}t\hat{n}\cdot\vec{\sigma}}, \quad \text{where} \quad \hat{H} = E_0\mathbb{1} + \frac{\hbar}{2}\vec{\Omega} \cdot \vec{\sigma}, \quad \vec{\Omega} = |\vec{\Omega}|\hat{n},$$

$$e^{-i\frac{|\vec{\Omega}|}{2}t\hat{n}\cdot\vec{\sigma}} \hookrightarrow \text{rotation of the Bloch vector by } \theta.$$

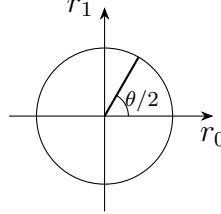
- $e^{-i\frac{\pi}{2}\sigma_x} = -i\sigma_x, \quad -i\sigma_x |g\rangle = -i|e\rangle \quad (\pi\text{-pulse about } x).$

2.3 The general form of a state

$$|\psi\rangle = c_0 |0\rangle + c_1 |1\rangle, \quad c_0, c_1 \in \mathbb{C}, \quad |c_0|^2 + |c_1|^2 = 1.$$

Let $c_0 = r_0 e^{i\alpha_0}$, $c_1 = r_1 e^{i\alpha_1}$; then

$$|c_0|^2 + |c_1|^2 = r_0^2 + r_1^2 = 1, \quad r_0, r_1 \geq 0 \quad \Rightarrow \quad r_0 = \cos\left(\frac{\theta}{2}\right), \quad r_1 = \sin\left(\frac{\theta}{2}\right), \quad \theta \in [0, \pi].$$



So

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + \underbrace{e^{i(\alpha_1 - \alpha_0)}}_{\text{global phase doesn't matter}} \sin\left(\frac{\theta}{2}\right) |1\rangle.$$

2.4 Expectation values of σ_x , σ_y , σ_z

$$A = \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix}, \quad |\psi\rangle = \begin{pmatrix} c_0 \\ c_1 \end{pmatrix},$$

$$\langle\psi|A|\psi\rangle = A_{00}|c_0|^2 + A_{01}c_0^*c_1 + A_{10}c_0c_1^* + A_{11}|c_1|^2.$$

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow \langle\sigma_z\rangle = |c_0|^2 - |c_1|^2 = \cos^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right),$$

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \langle\sigma_x\rangle = c_0^*c_1 + c_0c_1^* = c_0^*c_1 + (c_0^*c_1)^* = 2\operatorname{Re}(c_0^*c_1),$$

$$c_0^*c_1 = \cos\left(\frac{\theta}{2}\right)e^{i\varphi}\sin\left(\frac{\theta}{2}\right) = \frac{e^{i\varphi}}{2}\sin(\theta) \Rightarrow \langle\sigma_x\rangle = \sin(\theta)\cos(\varphi).$$

Similarly $\langle\sigma_y\rangle = 2\operatorname{Im}(c_0^*c_1) = \sin(\theta)\sin(\varphi)$. So

$$\vec{s} \equiv (\langle\sigma_x\rangle, \langle\sigma_y\rangle, \langle\sigma_z\rangle) = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$$